



**Hope Foundation's
International Institute of Information Technology, Pune**

#REF!

Academic Year 2026-27 Semester I

LIST OF THEORY ASSIGNMENTS

#REF!

Class: TE (2024) DIV-A

#REF!

Que No.	Question	Course Outcome	Marks	BTL
Unit 1 (CO304.1)				
Q1.	What are different objectives of AI?		5	BTL2
Q2.	Write a short note on : History of AI.		5	BTL2
Q3.	Define Rationality and rational agent. Give an example of rational action performed by any intelligent agent.		5	BTL2
Q4.	Define Intelligent Agent. What are the characteristics of Intelligent Agent.		5	BTL2
Q5.	Explain Model based Reflex agent with block diagram.		5	BTL2
Q6.	Explain Utility based agent with block diagram.		5	BTL2
Q7.	Define Artificial Intelligence, Intelligent Agent and its use.		5	BTL2
Q8.	What is an agent and its environment in AI? List the different types of environments with examples.		5	BTL2
Q9.	Write a note on the History of AI. Explain different applications of AI.		5	BTL2
Q10.	Describe Real life applications of AI.		5	BTL2
Unit 2 (CO304.2)				
Q1.	Briefly explain problem solving. Steps performed in problem-solving		5	BTL2
Q2.	Explain Bidirectional Search Algorithm and its Advantages disadvantages.		5	BTL2
Q3.	Explain with example various uninformed search techniques.		5	BTL3
Q4.	Compare uninformed and informed search techniques with suitable examples		5	BTL2
Q5.	What are the various problem characteristics for heuristic search?		5	BTL2
Q6.	Differentiate between Breadth-First Search (BFS) and Depth-First Search (DFS) with suitable examples.		5	BTL2
Q7.	Describe Hill climbing search with suitable example. List disadvantage of hill climbing process.		5	BTL3
Q8.	Explain A* algorithm in detail with a suitable example.		5	BTL3
Q9.	Describe the local search and optimization problem.		5	BTL3
Q10.	Compare and Contrast Depth First and Best First Search methods.		5	BTL4
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Signature of Course Teacher		Head of Department		

